

TROIS-RIVIERES, QC, Canada

WMO: 717240

Lat: 46.3537N

Lon: 72.5155W

Elev: 6

StdP: 101.25

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.8	-21.2	-29.2	0.3	-22.9	-26.4	0.3	-20.5	13.0	-2.3	11.4	-4.0	3.0	290	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.5	27.4	21.3	26.2	20.8	25.1	20.2	22.9	25.8	22.1	24.9	21.3	24.1	4.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.9	16.5	24.8	21.1	15.7	24.0	20.2	14.9	23.4	67.8	26.0	64.8	25.0	62.0	24.1	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.7	(b) 9.3	(c) 8.2	(e) -26.9	(f) 30.2	(g) 3.1	(h) 1.9	(i) -29.1	(j) 31.6	(k) -31.0	(l) 32.7	(m) -32.7	(n) 33.8	(o) -34.9	(p) 35.2		
(a) 10.7	(b) 9.3	(c) 8.2	(e) -27.0	(f) 24.3	(g) 3.0	(h) 1.1	(i) -29.2	(j) 25.1	(k) -30.9	(l) 25.7	(m) -32.6	(n) 26.3	(o) -34.7	(p) 27.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-10.0	-8.3	-2.9	4.5	12.5	18.0	20.8	19.9	15.7	8.8	1.7	-5.2
	DBStd	11.64	6.75	5.91	5.36	3.90	3.72	3.19	2.45	2.61	3.61	3.87	4.62	5.78
	HDD10.0	2507	619	513	399	168	17	0	0	0	2	69	249	471
	HDD18.3	4560	877	747	658	414	184	43	6	14	96	294	498	729
	CDD10.0	1188	0	0	0	4	94	241	335	308	172	33	1	0
	CDD18.3	199	0	0	0	0	3	34	82	63	15	1	0	0
	CDH23.3	748	0	0	0	0	13	121	333	237	44	0	0	0
	CDH26.7	73	0	0	0	0	2	13	38	17	3	0	0	0
Wind	WSAvg	3.8	3.9	4.0	4.0	3.9	3.9	3.5	3.6	3.4	3.5	3.8	3.9	3.8
Precipitation	PrecAvg	1014	72	56	68	78	85	97	106	99	95	95	81	87
	PrecMax	1189	133	117	121	164	138	205	199	195	184	183	139	176
	PrecMin	816	33	27	23	28	36	43	43	37	50	29	21	29
	PrecStd	107	27	21	25	34	29	35	37	31	32	36	26	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	4.7	9.5	17.7	25.0	28.2	29.5	28.4	26.6	21.1	14.5	6.8
		MCWB	3.8	3.0	5.3	10.9	16.8	21.1	21.5	21.8	21.3	16.5	12.2	5.6
	2%	DB	2.7	3.0	6.4	14.4	22.3	26.1	27.6	26.9	24.6	18.4	11.4	4.7
		MCWB	2.0	1.6	3.5	9.1	15.2	19.5	21.6	21.1	20.4	15.2	9.6	3.6
	5%	DB	1.4	1.8	4.8	11.9	20.3	24.7	26.4	25.8	22.9	16.3	9.4	3.3
		MCWB	0.7	0.8	2.4	7.5	14.7	18.9	21.1	20.8	19.1	13.4	7.5	2.3
	10%	DB	-0.2	0.5	3.4	10.0	18.4	23.1	25.2	24.6	21.4	14.4	7.7	2.1
		MCWB	-0.9	-0.4	1.5	6.3	13.4	18.2	20.6	20.2	18.0	11.9	6.1	1.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	3.6	6.5	12.7	19.4	22.6	24.2	23.9	22.5	17.9	12.7	6.0
		MCDB	4.4	4.1	7.7	15.7	23.2	26.2	27.5	27.0	25.4	19.9	13.6	6.5
	2%	WB	2.1	1.8	4.1	9.9	17.3	21.4	23.2	22.7	21.2	15.6	10.0	3.9
		MCDB	2.6	2.6	5.9	12.7	20.2	24.6	26.0	25.5	23.7	17.7	11.1	4.5
	5%	WB	0.8	0.9	2.8	8.3	15.8	20.4	22.3	21.8	20.0	13.8	7.8	2.5
		MCDB	1.3	1.7	4.2	11.2	18.8	23.3	25.0	24.5	22.4	15.8	9.0	3.2
	10%	WB	-0.9	-0.4	1.8	6.8	14.3	19.4	21.5	21.0	18.5	12.4	6.3	1.2
		MCDB	-0.2	0.5	3.1	9.3	17.6	22.0	24.3	23.7	20.9	14.2	7.6	1.9
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.4	7.6	7.6	8.5	7.9	7.5	7.7	7.9	6.4	5.4	5.9
		MCDBR	8.1	7.2	7.7	11.1	11.2	9.5	8.7	8.5	8.9	8.9	7.5	5.8
		MCWBR	8.1	6.7	5.6	7.1	5.9	5.2	4.9	5.1	5.8	6.4	6.7	5.6
	5% WB	MCDBR	7.8	7.0	7.1	10.1	9.4	8.1	7.6	7.6	8.1	8.1	7.3	5.6
		MCWBR	7.9	6.7	5.5	7.2	6.2	5.2	4.9	5.2	5.7	6.3	6.9	5.7
Clear-Sky Solar Irradiance	taub	0.262	0.285	0.318	0.350	0.374	0.414	0.417	0.407	0.374	0.336	0.308	0.279	
	taud	2.328	2.246	2.280	2.378	2.378	2.263	2.267	2.301	2.381	2.492	2.485	2.347	
	Ebn at Noon	844	888	907	907	892	854	845	839	837	826	786	774	
	Edh at Noon	74	99	112	112	117	132	130	121	102	79	65	66	
All-Sky Solar Radiation	RadAvg	1.42	2.31	3.61	4.49	5.13	5.62	5.55	4.86	3.81	2.23	1.33	0.96	
	RadStd	0.19	0.26	0.33	0.45	0.54	0.41	0.49	0.23	0.31	0.18	0.17	0.08	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page